Zhuang Zhuang (George) Han

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YouTube: www.youtube.com/@AIxbiology

EDUCATION

PhD in Medical Sciences, Cambridge Trust Scholar 2022.10 - Present
University of Cambridge, Cambridge, United Kingdom

Master of Science (Thesis-based), Biochemistry 2018.09 - 2021.08
University of Alberta, Edmonton, Canada

Bachelor of Science with Honours Biochemistry (First Class Honours) 2014.09 - 2018.06
University of Alberta, Edmonton, Canada

Certificate in Biomedical Research 2014.09 - 2018.06
University of Alberta, Edmonton, Canada

TECHNICAL SKILLS

Computational & data analysis

- Python for scientific computing, data analysis, and visualisation (NumPy, pandas, SciPy, matplotlib)
- Development of py-mea-axion, an open-source Python package for the analysis of multi-electrode array recordings, including burst detection, spike-timing tiling coefficient (STTC) synchrony analysis, and mixed-effects longitudinal statistics (https://www.piwheels.org/project/py-mea-axion/)
- Application of generative deep learning approaches (diffusion-based protein design) for de novo design of peptide binders against intrinsically disordered regions of transcription factors
- Currently developing formal foundations in machine learning through Andrew Ng's Machine Learning Specialization (Coursera, in progress)
- Open-source code sharing via GitHub (github.com/Molecularbiologyworld)

Neuronal cell culture & iPSC models

- Human iPSC maintenance, quality control, and differentiation into i3neurons
- Neuronal activity measurement using multi-electrode arrays (MEA)
- Modelling neurological disorders (GM3 synthase deficiency, LGI2 deficiency) in iPSC-derived neurons

Cell biology

- Engineering stable cell lines using lentiviral transduction and plasmid transfection
- Recombinant protein expression using HEK293T/F systems

Imaging & quantitative microscopy

- Confocal fluorescence microscopy and quantitative image analysis (e.g., ImageJ)
- Immunostaining (ICC) for neuronal and nuclear markers
- High-content screening microscopy for aggregate detection at scale

Biochemistry & proteomics

- Mass spectrometry-based proteomics (sample preparation and data interpretation)
- Western blotting, co-immunoprecipitation, protein interaction assays, and qRT-PCR
- Recombinant protein expression and purification (affinity & size-exclusion chromatography)
- Glycan profiling by HPLC
- Standard molecular biology: cloning, PCR, and gel electrophoresis

EMPLOYMENT HISTORY AND RESEARCH EXPERIENCE

BioSpark Company-Seeding Fellow 2025.11 - Present
Estel Bio, Cambridge
- Apply generative deep learning methods (diffusion-based protein design) to develop binders against intrinsically disordered transcription factor regions

BioSpark Challenge-Solving Fellow 2024.10 - 2025.07
Altos Labs, Cambridge
- Consult on chromatin ageing hallmarks, collaborating closely with principal investigators at the Altos Labs

PhD Student 2024.01 - Present
Deane Laboratory, Cambridge Institute for Medical Research, University of Cambridge
- Characterise interactions between GALC and TMEM106B in neurodegenerative diseases
- Model infantile-onset epilepsy with GM3 synthase deficient iPSC-derived neurons

PhD Student 2022.10 - 2023.12
Larrieu Laboratory, Department of Pharmacology, University of Cambridge
* I transferred to Professor Janet Deane's lab due to Dr. Larrieu's departure from Cambridge.
- Evaluate the benefit of new progeria rescue genes for other age-related pathologies
- Investigate nuclear envelope function and regulation in naked mole rat cells

Laboratory Technician 2022.03 - 2022.06
Goping Laboratory, Department of Biochemistry, University of Alberta
- Compound testing on breast cancer cell lines
- Compound testing on mouse/patient-derived breast cancer organoids

Laboratory Technician 2021.09 - 2022.02
Westaway Laboratory, Centre for Prions and Protein Folding Diseases (CPPFD)
Department of Medicine, University of Alberta
- High-content screening for compounds that modulate tau aggregation kinetics in biosensor cells

Graduate Research Assistant (MSc. Student) 2018.09 - 2021.08
Westaway Laboratory, Centre for Prions and Protein Folding Diseases (CPPFD)
Department of Biochemistry, University of Alberta
- Introduce progeric mutations to Alzheimer's disease cell culture and animal models to better mimic the biochemical environment of an aged central nervous system
- Investigate the early aging-associated molecular events in the development of Alzheimer's disease

Undergraduate Researcher (BSc. Student) 2015.05 - 2018.08
Schultz Laboratory
Department of Biochemistry, University of Alberta
- Study enzyme activity of nanoparticle-enzyme complex in blood
- Study proteomes of nanoparticle-adsorbed cytosolic proteins

Undergraduate Researcher (BSc. Student) 2017.07 - 2017.08
Jackson Laboratory
Deutsches Zentrum für Neurodegenerative Erkrankungen (DZNE), Bonn, Germany
- Evaluate and improve the thiouracil-tagging method for brain cell type-specific RNA labeling

PUBLICATIONS

Peer-reviewed publications

- Barrow H*, **Han ZZ***, Nicholson AS, Strasser S, Nash DA, Suberu JO, Antrobus R, te Vruchte D, Priestman DA, Graham SC, Platt FM, Deane JE 2026.03
  Human iPSC Models of Ganglioside Deficiency Reveal a Sialylated Lipid Requirement for Plasma-Membrane Organization and Neuronal Activity.
  BioRxiv. *doi.org/10.64898/2026.03.18.712603*
  * Equal first author

- **Han ZZ**, de Magalhães JP. 2026.03
  The Ethics Case for Longevity Science.
  Ageing Research Reviews. *doi: 10.1016/j.arr.2026.103054*

- **Han ZZ**, Kang SG, Gomez-Cardona E, Wohlgemuth S, Shmeit K, Arce LA, Safar JG, Julien O, Westaway D. 2025.07
  Expression of progerin enhances disease-related endpoints in a tau seeding reporter cell system.
  GeroScience. doi.org/10.1007/s11357-025-01737-z

- Drugachenok P, Urriola-Muñoz P, Qiu L, **Han ZZ**, St John Smith E, Rahman T. 2025.06
  Functional characterization of the store-operated calcium entry pathway in naked mole-rat cells.
  Open Biology. doi.org/10.1098/rsob.250052

- Githaka JM, Gutiérrez T, **Han ZZ**, Primeau JO, Muranyi H, Young HS, Sykes BD, Julien O, Goping IS. 2025.05
  The non-cytotoxic small molecule NPB does not inhibit BAD phosphorylation and forms colloidal aggregates.
  Communications Medicine. doi.org/10.1038/s43856-022-00142-3

- **Han ZZ**, Fleet A, Larrieu D. 2023.04
  Can accelerated ageing models inform us on age-related tauopathies?
  Aging Cell. doi: 10.1111/acel.13830.

- **Han ZZ**, Kang SG, Arce LA, Westaway D. 2022.04
  Prion-like strain effects in tauopathies.
  Cell and Tissue Research. doi.org/10.1007/s00441-022-03620-1

- **Han ZZ**, Schultz MC. 2022.02
  The protein interactome of a nanoparticle population in whole cytoplasm under near-native conditions: a pilot study.
  Particle & Particle Systems Characterization. doi.org/10.1002/ppsc.202100283

- Kim C, Haldiman T, Kang SG, Hromadkova L, **Han ZZ**, Chen W, Lissemore F, Lerner A, de Silva R, Cohen ML, Westaway D, Safar JG. 2022.01
  Distinct populations of highly potent tau seed conformers in rapidly progressing Alzheimer's disease.
  Science Translational Medicine. doi: 10.1126/scitranslmed.abg0253

- Kang SG, **Han ZZ**, Daude N, McNamara E, Wohlgemuth S, Molina-Porcel L, Safar JG, Mok SA, Westaway D. 2021.09
  Pathologic tau conformer ensembles induce dynamic, liquid-liquid phase separation events at the nuclear envelope.
  BMC Biology. doi: 10.1186/s12915-021-01132-y

- Daude N, Kim C, Kang SG, Eskandari-Sedighi G, Haldiman T, Yang J, Fleck SC, Gomez-Cardona, **Han ZZ**, Borrego-Ecija S, Wohlgemuth S, Julien O, Wille H, Molina-Porcel L, Gelpi E, Safar JG, Westaway D. (2020) 2020.03
  Diverse, evolving conformer populations drive distinct phenotypes in frontotemporal lobar degeneration caused by the same MAPT-P301L mutation.
  Acta Neuropathologica. doi: 10.1007/s00401-020-02148-4

Conference presentations

- *Unravelling the effects of GM3 synthase deficiency in human iPSC-derived neurons* 2025.09
  8th Cambridge Neuroscience Symposium, Cambridge

- *Expression of progerin enhances disease-related endpoints in a tau seeding reporter cell system* 2025.08
  The 12th Aging Research & Drug Discovery Meeting, Copenhagen

- *Unravelling the effects of GM3 synthase deficiency in human iPSC-derived neurons* 2025.05
  Wolfson Research Event 2025, Cambridge

- *Structural characterisation of GALC-TMEM106B complex* 2024.09
  Cambridge Institute for Medical Research Retreat 2024, Cambridge

- *Effects of lamin A progeric mutation in tau reporter cell lines* 2021.05
  Promoting Healthy Brain Aging and Preventing Dementia Symposium 2021, Edmonton

- *Investigation of the role of Lamin A-induced ageing in tauopathies.* 2019.05
  Poster presentation at the PRION 2019, Edmonton
  PRION 2019 emerging concepts, Prion, 13:sup1, 1-141

- *Gel electrophoresis and membrane blotting for analyzing noncovalent interactions between nanoparticles and proteins.* 2016.11
  Poster presentation at the Department of Biochemistry Retreat 2016, Banff

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AWARDS AND HONOURS

Wolfson College Travel & Research Grant	2025
2nd place in AI in Longevity Hackathon (organised by GetSeen ventures and London Longevity)	2025
Early Career Researcher Spinout Idea Competition Prize – 73rd Annual Scientific Meeting of the British Society for Research on Ageing (BSRA)	2024
Cambridge International Scholarship (Cambridge Trust)	2022
CIHR Covid Extensions	2020
Gyro Club of Edmonton Graduate Scholarship – Master's level	2020
Best Project Summary – CIHR Institute of Aging Summer Program in Aging 2019 & Quebec Network for Research on Aging	2019
CIHR Travel Awards – Institute Community Support	2019
Walter H Johns Graduate Fellowship	2019
Frederick Banting and Charles Best Canada Graduate Scholarship – Master's	2019
Lionel B Pett Graduate Scholarship	2018
Queen Elizabeth II Graduate Scholarship – Master's level	2018
University of Alberta Thesis-Based Master's Recruitment Scholarship	2018
Hooper-Munroe Academic Award	2018
Morgan-Scraba Prize	2018
Dean's Silver Medal in Science	2018
Office of the Provost and VP (Academic) Summer Student Award	2018
Dean's Honor Roll	2018
Jason Lang Scholarship	2017

James McCrie Douglas Memorial Scholarship	2017
Bill Paranchych Memorial Scholarship	2017
Education Abroad Individual Award	2017
Dean's Honor Roll	2017
Coca Cola Student Achievement Award	2016
Jason Lang Scholarship	2016
Dean's Honor Roll	2016
Jason Lang Scholarship	2015
Dean's Honor Roll	2015
University of Alberta Academic Excellence Scholarship	2014
Faculty of Science Academic Excellence Scholarship	2014

PROFESSIONAL TRAININGS

Summer Program in Aging (SPA) 2019.05
The CIHR Institute of Aging, Estérel Resort, Québec

Graduate Ethics Training 2018.10
University of Alberta, Edmonton

Surgical Training 2018.06
University of Alberta, Edmonton

- Certificate: Introduction to Basic Surgical Techniques in Research
- Certificate: Mouse Isoflurane Anesthesia Subcutaneous, and Intraperitoneal Injections
- Certificate: Aseptic Techniques in Surgery

LEADERSHIP EXPERIENCE

Academic Tutor (Biology) 2024.07 - 2025.08
Immerse Education Services

President 2024.06 - 2025.09
Cambridge Society of Longevity and Ageing Research

Treasurer 2023.01 - 2024.05
Cambridge Society of Longevity and Ageing Research

Chair 2023.10 - 2024.05
Wolfson Research Event 2024, Wolfson College, University of Cambridge

Editorial Officer 2022.10 - 2023.05
Wolfson Research Event 2023, Wolfson College, University of Cambridge

Vice President of Social 2020.05 - 2021.06
Biochemistry Graduate Students' Association (BCGSA), University of Alberta

Director of Education/Advocacy 2019.10 - 2020.05
Alzheimer's Students Association (ASA), University of Alberta

Vice President of External 2019.05 - 2020.05
Biochemistry Graduate Students' Association (BCGSA), University of Alberta

President 2017.09 - 2018.05
Biochemistry Students' Association (BCSA), University of Alberta

Vice President of Communications 2016.09 - 2017.04
Biochemistry Students' Association (BCSA), University of Alberta